Lecture 8: Confidence Intervals & the Bootstrap

Two Paths to Quantifying Uncertainty

Previously, on Lecture 7...

Sampling distribution: $\hat{\theta}$ is random — different samples give different estimates.

CLT: $\bar{X} \dot{\sim} N(\mu, \sigma^2/n)$. The MLE is also asymptotically Normal: $\hat{\theta} \dot{\sim} N(\theta, 1/(nI(\theta)))$.

Standard error: $\text{SE} = \text{SD}(\hat{\theta})$. For $\bar{X}$: $\text{SE} = \sigma/\sqrt{n}$. For any MLE: $\text{SE} \approx 1/\sqrt{nI(\hat{\theta})}$.

$\sqrt{n}$ **law:** Halving the SE requires quadrupling n. Precision is expensive.

SD $\neq$ SE: SD measures data spread (fixed). SE measures estimator precision (shrinks with n).

Today: Two ways to **use** the sampling distribution.

"52% support candidate A" means nothing without $\pm$ something.

Part I: Confidence Intervals

The analytical path: formulas from theory

What Is a Confidence Interval?

A **95% confidence interval** is a random interval $[L, U]$ such that

$$P(L \leq \theta \leq U) = 0.95$$

Before you collect data, there is a 95% chance the interval will contain θ.

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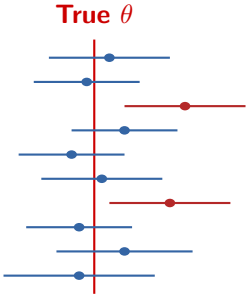

10 simulated 95% CIs

2 miss θ (expected $\approx$ 1 in 20)

Constructing the Wald CI

Since $\hat{\theta}$ is approximately Normal (by CLT or MLE asymptotics): $\frac{\hat{\theta}-\theta}{\mathrm{SE}(\hat{\theta})} \approx N(0,1)$

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$$\hat{\theta} \quad \pm \quad z_{\alpha/2} \cdot \mathrm{SE}(\hat{\theta})$$

For 95%: $\hat{\theta} \pm 1.96 \cdot \mathrm{SE}$ For 99%: $\hat{\theta} \pm 2.576 \cdot \mathrm{SE}$ For 90%: $\hat{\theta} \pm 1.645 \cdot \mathrm{SE}$

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Recipe:

1. Compute the point estimate $\hat{\theta}$
2. Compute (or estimate) the standard error $\mathrm{SE}(\hat{\theta})$
3. Choose the confidence level $\Rightarrow$ look up $z_{\alpha/2}$
4. Report: $\hat{\theta} \pm z_{\alpha/2} \cdot \mathrm{SE}$

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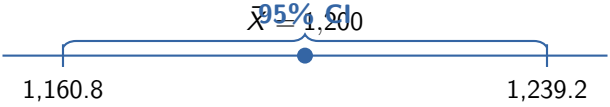
Step 4:

$$\bar{X} \pm 1.96 \cdot \text{SE} = 1{,}200 \pm 1.96 \times 20 = 1{,}200 \pm 39.2$$



"We are 95% confident that the true mean lifetime is between 1,161 and 1,239 hours."

What 95% Confidence **Really** Means

Correct:

If we repeated the experiment many times, 95% of the resulting intervals would contain θ.

The *procedure* works 95% of the time.

Analogy: A 95%-accurate archer hits the target 95% of the time. After the arrow lands, it either hit or missed — the probability is gone.

A CI is the arrow. Once computed, it either contains θ or it doesn't.

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What Determines the Width?

$$\text{Width} = 2 \cdot z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

Confidence level $\uparrow$

$z_{\alpha/2}$ increases

$\Rightarrow$ **wider** CI

90%: 1.645
95%: 1.960
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Sample size n $\uparrow$

$\sqrt{n}$ in denominator

$\Rightarrow$ **narrower** CI

But diminishing returns ($\sqrt{n}$ law!)

CI for a Proportion: Wald vs Wilson

Setup: $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$. MLE: $\hat{p} = k/n$. SE $= \sqrt{\hat{p}(1-\hat{p})/n}$.

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Example: $n = 1{,}000$, $\hat{p} = 0.52 \Rightarrow [0.489, 0.551]$.

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Wilson interval — adds "pseudo-observations" (like MAP!):

$$\tilde{p} = \frac{k+z^2/2}{n+z^2} \qquad \text{Wilson CI: } \tilde{p} \pm \frac{z}{n+z^2}\sqrt{k(n-k)/n + z^2/4}$$

For 95%: acts like adding 2 successes and 2 failures $\Rightarrow \tilde{p} \approx (k+2)/(n+4)$.

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Rule of thumb: Use Wald when $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$.
Otherwise use Wilson. Many stats packages default to Wilson.

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$\text{ME} = \pm 5\%$ $n = 385$	$\text{ME} = \pm 3\%$ $n = 1{,}068$	$\text{ME} = \pm 1\%$ $n = 9{,}604$

The $\sqrt{n}$ law again: $10\times$ more precision requires $100\times$ more data.

The t-Interval and the General MLE Recipe

When σ is unknown (the typical case), use the t-distribution:

$$\bar{X} \quad \pm \quad t_{n-1,\,\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

t_{n-1} has heavier tails than $N(0,1) \Rightarrow$ wider CI. For $n \geq 30$, nearly identical to z.

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Example: Poisson λ. $I(\lambda) = 1/\lambda$, MLE $= \hat{\lambda} = \bar{X}$. SE $= \sqrt{\hat{\lambda}/n}$.
$n = 50$ earthquakes/year, $\hat{\lambda} = 3.2$: 95% CI $= 3.2 \pm 1.96\sqrt{3.2/50} = [2.70, 3.70]$.

Confidence Intervals vs Credible Intervals

Confidence Interval (Frequentist)

"95% of intervals built this way contain θ"

θ is **fixed**; the interval is random.

No prior needed.

$\hat{\theta} \pm z \cdot \text{SE}$

With large n, they often give nearly identical intervals.
The philosophical difference matters most with small n or strong priors.

Neither is "wrong" — they answer **different questions**.

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Example: Odds ratio. From $\hat{p}$, compute $\hat{\text{OR}} = \hat{p}/(1-\hat{p})$.

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Tip: Often easier to build CI on the log scale, then exponentiate the endpoints.

Part II: The Bootstrap

The computational path: let the data speak

When Formulas Don't Exist

Part I world:

Mean, proportion, MLE
SE formulas exist
Normal approx-
imation works

→

Real world:

Median, correlation, ratio
SE formula is
hard or unknown
Distribution
may be skewed

→

Bootstrap solution:

Let the *data itself*
tell you the SE
No formulas needed

The Core Insight

Ideal world

Draw many samples from the *true* population F

Compute $\hat{\theta}$ on each

$\Rightarrow$ exact sampling distribution

We can't do this

so we do this instead

Bootstrap world

Draw many samples from the *observed data* $\hat{F}_n$

Compute $\hat{\theta}^*$ on each

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Key idea: The empirical distribution $\hat{F}_n$ is our best estimate of F. Sampling from $\hat{F}_n$ = sampling **with replacement** from the data.

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Step 1: Start with your original data $\mathbf{x} = (x_1, x_2, \ldots, x_n)$.

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Step 5: Use the distribution of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ to estimate SE, bias, or CI.

Visualizing a Bootstrap Sample

Original data: $\mathbf{x} = (2, 5, 7, 3, 8, 1, 6), \quad n = 7.$

Original:

2	**5**	**7**	**3**	**8**	**1**	**6**

Note: some values appear **multiple times**, others **not at all**. That's "with replacement."

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Boot #1:	**5**	**3**	**3**	**8**	**2**	**5**	**7**

$\hat{\theta}_1^* = \text{median} = 5$

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Boot #2:	**7**	**7**	**1**	**6**	**8**	**3**	**8**	$\hat{\theta}_2^* = \text{median} = 7$

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Boot #1:	5	3	3	8	2	5	7	$\hat{\theta}_1^* = \text{median} = 5$
Boot #2:	7	7	1	6	8	3	8	$\hat{\theta}_2^* = \text{median} = 7$
Boot #3:	2	1	6	2	3	1	5	$\hat{\theta}_3^* = \text{median} = 2$

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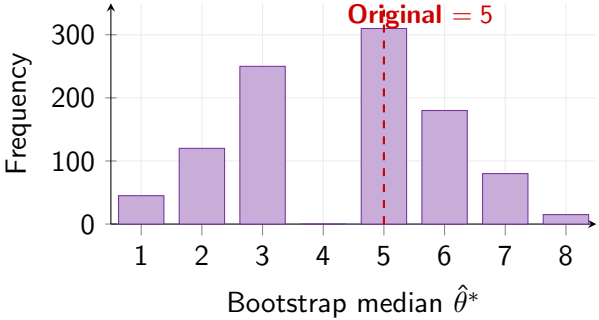
Bootstrap SE and the Bootstrap Distribution



$$\widehat{\mathrm{SE}}_{\mathrm{boot}} = \mathrm{SD}(\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*)$$

The **spread** of the bootstrap distribution estimates the SE.

No formula needed — just simulation!

How many replicates? $B \geq 200$ for SE, $B \geq 1{,}000$ for percentile CI, $B \geq 5{,}000$ for BCa CI.

Bootstrap Confidence Intervals: Three Methods

Normal

$$\hat{\theta} \pm z_{\alpha/2} \cdot \widehat{\mathsf{SE}}_{\text{boot}}$$

Same as Wald, but with bootstrap SE instead of formula.

Assumes symmetry.

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$$[\hat{\theta}^*_{(0.025)},\ \ \hat{\theta}^*_{(0.975)}]$$

Just take the 2.5th and 97.5th percentiles of the bootstrap dist.

Respects asymmetry.

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BCa

Adjusted percentiles

Corrects for bias and skewness. Best coverage.

Gold standard. Use by default.

$\longrightarrow$ increasing sophistication and accuracy

Why Does the Bootstrap Work?

Real world

$$F \xrightarrow{\text{sample}} \mathbf{x} \xrightarrow{T} \hat{\theta}$$

Variation of $\hat{\theta}$ around θ
= sampling distribution

analogous

Bootstrap world

$$\hat{F}_n \xrightarrow{\text{resample}} \mathbf{x}^* \xrightarrow{T} \hat{\theta}^*$$

Variation of $\hat{\theta}^*$ around $\hat{\theta}$
= bootstrap distribution

Bootstrap principle: The relationship
$\hat{F}_n \to \hat{\theta}^*$ *mimics* the relationship $F \to \hat{\theta}$.
Formally: $\hat{\theta}^* - \hat{\theta} \approx_d \hat{\theta} - \theta$ as $n \to \infty$.

Parametric vs Nonparametric Bootstrap

Nonparametric

Resample **with replacement** from the original data

No distributional assumptions

$$x_1^*, \ldots, x_n^* \sim \hat{F}_n$$

Default choice

Parametric

Fit a model, then simulate **new data** from the fit

Assumes a distribution (e.g., Normal)

$$x_1^*, \ldots, x_n^* \sim f(\cdot \mid \hat{\theta})$$

More efficient if model is correct

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$$x_1^*, \ldots, x_n^* \sim \hat{F}_n$$

Default choice

Parametric

Fit a model, then simulate **new data** from the fit

Assumes a distribution (e.g., Normal)

$$x_1^*, \ldots, x_n^* \sim f(\cdot \mid \hat{\theta})$$

More efficient if model is correct

Rule of thumb: Use nonparametric unless you have strong reasons to trust the model.
Important: For paired data (e.g., correlation), resample *pairs*, not individual values!

When the Bootstrap Fails

1. Extremes and tails

Max, min, extreme quantiles. Bootstrap can't generate values outside the observed range.

Classic example: $X_i \sim \text{Uniform}(0, \theta)$, $\hat{\theta} = X_{(n)}$. The bootstrap can *never* exceed $X_{(n)}$, so it underestimates variability. (See Lecture 5: MLE for Uniform.)

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4. Non-smooth statistics

Mode, number of clusters. Statistics that change discontinuously.

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Two Paths, One Goal

When to use which?

Analytical CI vs Bootstrap CI

Analytical (Wald)

$\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}$

+ Fast, no simulation
+ Exact SE formulas for common cases
+ Easy to interpret

– Needs a formula for SE
– Assumes approximate normality
– Always symmetric

Bootstrap

Resample $\Rightarrow$ percentiles

+ Works for *any* statistic
+ No formulas needed
+ Respects asymmetry (percentile/BCa)

– Computationally expensive
– Fails for extremes, small n
– Results vary across runs

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– Fails for extremes, small n
– Results vary across runs

In practice: Use analytical CIs when formulas exist and assumptions hold.
Use bootstrap when: the statistic is complex (median, correlation, ratio),
no SE formula is available, or you suspect the distribution is skewed.

Bootstrap in Python: It's Simple

```python
import numpy as np
from scipy import stats

# Original data
x = np.array([2, 5, 7, 3, 8, 1, 6])
n = len(x)

# Nonparametric bootstrap
B = 10000
boot_medians = np.array([
  np.median(np.random.choice(x, size=n, replace=True))
  for _ in range(B)
])

# Bootstrap SE and percentile CI
se_boot = np.std(boot_medians, ddof=1)
ci_pct = np.percentile(boot_medians, [2.5, 97.5])

# For BCa: scipy.stats.bootstrap() since SciPy 1.7
res = stats.bootstrap((x,), np.median, n_resamples=B)
ci_bca = res.confidence_interval
```

Summary: Quantifying Uncertainty

Confidence interval: A random interval that contains θ with probability $1 - \alpha$. It's the *procedure* t

Wald CI: $\hat{\theta} \pm z_{\alpha/2} \cdot \text{SE}$. Uses normal approximation. Need SE formula.

Proportions: Use Wilson (not Wald) for small n or extreme $\hat{p}$.

t-interval: Use when σ is unknown and n is small. Heavier tails $\Rightarrow$ wider CI.

Bootstrap: Resample with replacement $\Rightarrow$ SE and CI without formulas.

Bootstrap CIs: Normal (simple), Percentile (respects shape), BCa (gold standard).

When bootstrap fails: Extremes, tiny n, dependent data, non-smooth statistics.

Two paths: Analytical when formulas exist; bootstrap when they don't. Same goal: quantify unc

Practical: Confidence Intervals & Bootstrap

1. **CI coverage simulation:**
   - Generate $n = 30$ from $N(\mu, \sigma^2)$, compute 95% Wald CI
   - Repeat 10,000 times. What fraction contain μ? (Should be $\approx 95\%$)
   - Try $n = 5$. Still 95%? Now try the t-interval. Better?
2. **Bootstrap vs analytical:**
   - Generate $n = 30$ from Exp(1), compute the median
   - Analytical: no easy formula! Bootstrap: compute $B = 5{,}000$ medians
   - Build percentile and BCa CIs. Does the true median ($\ln 2$) fall in?
3. **Wald vs Wilson for proportions:**
   - Simulate $n = 20$, $p = 0.05$. Compare coverage of Wald and Wilson
   - At what n does Wald catch up to Wilson's coverage?

Homework

1. A factory tests $n = 100$ lightbulbs: $\bar{X} = 1{,}150$ hours, $S = 200$ hours. Construct 90%, 95%, and 99% CIs for the true mean. Which is widest? Why?

2. A poll surveys $n = 500$ voters: 265 support candidate A ($\hat{p} = 0.53$).
   (a) Compute the Wald 95% CI. Can we declare A is leading?
   (b) How many voters do we need for a $\pm 2\%$ margin of error?

3. The heights (cm) of 12 students: 165, 170, 168, 172, 175, 180, 163, 178, 169, 171, 167, 174.
   (a) Compute the bootstrap SE of the **median** using $B = 5{,}000$.
   (b) Construct 95% Normal and Percentile bootstrap CIs.
   (c) Is the bootstrap distribution symmetric? Compare with the analytical CI for the mean.

4. For $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$: MLE $\hat{\lambda} = 1/\bar{X}$, $I(\lambda) = 1/\lambda^2$.
   (a) Build the Wald 95% CI for λ using Fisher information.
   (b) Use the delta method to find a CI for the mean $\mu = 1/\lambda$.
   (c) Compare with a bootstrap percentile CI for λ. Do they agree?

Recommended Visualizations & Resources

Interactive: Confidence Interval Simulation (R Psychologist)

`rpsychologist.com/d3/ci` — watch CIs accumulate in real time. Drag sliders for n, σ, confidence level. The best CI demo.

Interactive: Seeing Theory — Frequentist Inference (Brown)

`seeing-theory.brown.edu/frequentist-inference` — CI construction and bootstrap resampling animated step by step.

Video: StatQuest — Confidence Intervals & Bootstrapping

Two clear walkthroughs: what CIs are and are not, and how bootstrap builds CIs from data.

Reading: Efron & Tibshirani, "An Introduction to the Bootstrap"

The foundational textbook. Chapters 1–6 cover everything in Part II. Chapters 12–14 cover BCa.

Python: scipy.stats.bootstrap()

`docs.scipy.org/doc/scipy/reference/generated/scipy.stats.bootstrap.html` — BCa intervals in three lines of code (SciPy $\geq$ 1.7).

Questions?

Next: Lecture 9 — Hypothesis testing, p-values, power, and permutation tests